Advanced Number Theory

Section 1: Modular Arithmetic & Congruence

1 What is the remainder when 3^{42} is divided by 7?

a. 1

b. 2

c. 4

d. 6

2 Solve the linear congruence for x: $4x \equiv 3 \pmod 7$ with $0 \le x < 7$.

a. 2

b. 4

c. 5

d. 6

3 What is the units digit of the integer 7^{2024}?

a. 1

b. 3

c. 7

d. 9

Section 2: Diophantine Equations & Divisibility

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4 How many positive integer pairs (x, y) satisfy the linear Diophantine equation $3x + 5y = 105$?

a. 5

b. 6

c. 7

d. 8

5 If a positive integer N leaves a remainder of 2 when divided by 5 and a remainder of 3 when divided by 7, what is the smallest such positive integer?

a. 17

b. 24

c. 31

d. 38

6 For what value of the digit d is the six-digit integer $452,d76$ divisible by 11?

a. 1

b. 3

c. 5

d. 7

Section 3: Rational vs. Irrational & Continued Fractions

7 Express the repeating decimal $0.2\overline{45}$ as a simplified common fraction.

a. $\frac{27}{110}$

b. $\frac{49}{200}$

c. $\frac{1}{4}$

d. $\frac{12}{55}$

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8 Which of the following real numbers is irrational?

a. $\sqrt{0.49}$

b. $\sqrt[3]{-27}$

c. $\sqrt{18} - \sqrt{2}$

d. $\frac{\sqrt{72}}{\sqrt{2}}$

9 What positive real number is represented by the infinite continued fraction $x = 1 + \frac{1}{1 + \frac{1}{1 + \cdots}}$?

a. $\frac{1+\sqrt{3}}{2}$

b. $\frac{1+\sqrt{5}}{2}$

c. $\sqrt{2}$

d. $\frac{3}{2}$

Section 4: Base Conversions & Combinatorial Number Theory

10 Convert the base-5 numeral 342_5 to standard base-10 decimal notation.

a. 87

b. 97

c. 102

d. 112

11 How many trailing zeros does the decimal expansion of $50!$ contain?

a. 10

b. 12

c. 14

d. 16

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12 How many positive factors does the integer 720 have?

a. 24

b. 28

c. 30

d. 36

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